

Lim Chien Her

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EDUCATION

NANYANG TECHNOLOGICAL UNIVERSITY, SINGAPORE

Aug 2021 – May 2025

Bachelor of Science (Honours), Data Science and Artificial Intelligence

- Honours (Distinction) | CGPA 4.23/5
- Relevant Coursework – Introduction to Data Science and Artificial Intelligence, Data Analysis with Computer, Data Analytics and Mining, Data Visualisation

SKILLS

- Software Skills: Python, R, SQL, C#, Pandas, Tensorflow, Pytorch, Seaborn, Matplotlib, Numpy, Plotly, Tableau, Google Looker Studio, AWS, Unity
- IT Skills: Microsoft Office (Word, Excel, Powerpoint, Outlook & Teams), Davinci Resolve, Adobe (Photoshop & Illustrator)
- Transferrable Skills: Team work, Problem Solving, Analytical, Communication, Detail-oriented
- Language: Proficient in English & Chinese

WORK EXPERIENCE | INTERSHIP

MIAO WORLDS

Aug 2025 – Current

AI Engineer

Zero-Shot 3D Map Generation with LLM Agents: A Dual-Agent Architecture for PCG

- First-authored a research paper (arXiv: [2512.10501](https://arxiv.org/abs/2512.10501)) proposing a training-free dual-agent (Actor-Critic) architecture for zero-shot 3D procedural content generation
- Eliminated the need for fine-tuning or reinforcement learning used by prior methods and achieved an 80% task success rate with a 20% performance boost over single-agent baselines.

Comparative Benchmarking of Autonomous IDE Agents in Unity | *In Preparation*

- Led SOTA Benchmarking for Agentic Systems: Architected a reproducible Unity benchmarking framework (74 tasks across 6 categories) using an isolated Task Loader to prevent reward-hacking
- Conducted head-to-head comparisons of the company's AI Assistants vs. Claude Code (Sonnet 4.5/4.6) via Model Context Protocol (MCP), isolating framework-level performance gains

Full-Stack Retrieval-Augmented Generation (RAG) Infrastructure

- Implemented two-stage retrieval (FAISS vector search, cross encoder re-ranking), tree-sitter AST indexing for structural code understanding, and a Merkle-tree incremental sync engine for low-latency codebase ingestion
- Deployed as a GPU-accelerated FastAPI service with MongoDB and Docker, and exposed retrieval as an agent-callable tool, wiring semantic search into the LLM's tool-use loop to reduce hallucinated API calls

PUBLIC SERVICE DIVISION – PRIME MINISTER'S OFFICE

Dec 2023 – Jul 2024

Data Science Intern

- Architected an end-to-end data pipeline ingesting thousands of unstructured text responses from multiple agencies; cleaned, processed, and vectorized the data using Python to prepare it for ML and LLM analysis
- Engineered a novel NLP workflow by fine-tuning a Claude LLM model and Amazon SageMaker for sentiment and multi-label feature extraction, automating a previously manual analysis process and reducing manual effort by 20+ hours per month
- Leveraged advanced prompt engineering to transform the LLM into an "explainability engine", generating clear natural language rationales for its assessments, which enabled agencies to refine their communication and decrease repeated monthly appeals by 5%
- Synthesized processed data into actionable intelligence by building interactive Tableau dashboards uncovering critical bottlenecks that reduced average resolution time by 14 days

NTUC HEALTH

May 2023 – July 2023

Data Project Intern

- Cooperated with data team to develop and implement a centralised AWS database using SQL to streamline data storage, management and access, enhancing operational efficiency by 20%
- Designed visually engaging and informative dashboards using Google Looker Studio to enable seamless data visualisation and in-depth analysis, improving resource allocation by \$10,000/month

NATIONAL SERVICE

Apr 2019 – Feb 2021

3SIR/Combat Signaller

- Worked together with unit medic and doctor to establish communication from the battlefield to headquarters and manage casualties
- Achieved unit best platoon together with platoon mates for efforts in unit ATEC evaluation

PROJECT WORK

Final Year Project – Teaching Agents to play Table Tennis

Aug 2024 – May 2025

- Objective – Uncover novel methods to improve training efficiency and results
- Planned and executed a 12-month project roadmap, coordinating regularly with supervisor and project manager to align milestones, model iterations, and evaluation benchmarks
- Implemented SAC (Soft Actor Critic) to train Reinforcement Learning agents to play table tennis in a Unity ML-Agents environment, under the supervision of Prof. Seah Hock Soon (NTU)
- Designed a custom curriculum learning strategy that reduced training time by 50% while maintaining baseline performance of 20-long shot rallies
- Led experimentation and iterative tuning of dynamic hyperparameters, enabling agents to consistently perform advanced backhand top-spin rallies—achieving performance on par with professional players; awarded an A- grade

Hackathon – Kaggle Competition: Elo Merchant Recommendation

Aug 2023 – Dec 2023

- Objective – Predict customer loyalty for Elo (Brazilian payment brand) using transaction data.
- Optimized memory usage by refining data types and implementing a custom garbage collector.
- Implemented Feature Engineering by extracting temporal trends and holiday-season activity patterns to enhance predictive power.
- Optimized model by training a gradient boosting machine (GBM) on outlier/non-outlier splits, leveraging feature importance (gain/split metrics) to boost efficiency; achieved RMSE 3.6, ranking 914/4111 (top 22.2%)

Data Visualisation – Data Analysis on Billionaires in the World

Aug 2023 – Dec 2023

- Conducted exploratory data analysis (EDA) on Billionaire dataset from Kaggle using Seaborn, Matplotlib and Python to uncover trends in wealth distribution, industries and demographics
- Designed interactive dashboards (Plotly Dash) to dynamically filter data by country, industry, and wealth source
- Cleaned and preprocessed data by handling missing values, removing duplicates, and standardizing columns for accurate analysis